

# Developments in Mixed-Signal Oscilloscopes Accelerate Embedded Design Development



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Engineers, especially those working on embedded designs, bank on mixed-signal oscilloscopes (MSOs) right from the prototype stage to product testing. Embedded system designers debug their designs with analogue- and digital-channel capabilities in these oscilloscopes, which also provide a variety of triggering options. Let us take a look at the new capabilities added in these MSOs.

## Faster, wider, higher, deeper: four buzzwords in MSOs

As embedded designs continue to grow speedily in numbers, manufacturers are continuously evolving their MSOs accordingly. With new applications emerging in the Internet of Things (IoT), wireless and various semiconductor technologies, MSOs are playing an important tool for debugging.

Today's oscilloscopes come with faster sampling rates, wider channel selection, higher resolutions and deep memory.

For instance, HDO8000 series introduced by Teledyne LeCroy has eight analogue input channels with 12-bit resolution. This allows engineers to view clean and crisp signals in greater detail, post debugging using digital channels. Often signals are lost in noise. But with high resolution, engineers can be sure of clearly seeing such signals, thereby improving the overall debug and analysis process.

RTO and RTE series of digital-storage oscilloscopes (DSOs) from Rohde & Schwarz, which also have the option of mixed-signal, provide high-resolution capabilities with their monolithic analogue-to-digital (A/D) converters that make acquisitions up to 16-bit resolution possible.

Some mixed-signal oscilloscopes available in the market

